2023 International VLSI Symposium on Technology, Systems and Applications (VLSI-TSA/VLSI-DAT) | 979-8-3503-3416-6/23/$31.00 ©2023 IEEE | DOI: 10.1109/VLSI-TSA/VLSI-DAT57221.2023.10134000

Opportunities and Challenges of 2.5D Packaging in the AI Data Center Era

Shang Y. Hou
TSMC

The advent of 2.5D packaging integration technology in the last decade realized a performance breakthrough. This new class of packaging technology boosts the computing power greatly and develops into a standard in today’s fastest HPC and AI systems. A general performance index to gauge the performance is the total number of transistors within a package. Typically a few tens billion of logic transistors and memory transistors ~10x of logic. In this talk, I will list the key attributes that are required to increase the performance further at a higher power efficiency. These include larger interposer area, more global and semi-global interconnect layers, passives, and thermal solutions. The opportunities and challenges to develop such a system will also be addressed.